DATABASE DESIGN DOCUMENT

PART-1: DATABASE FOUNDATION

School ERP

PostgreSQL | Spring Data JPA | Hibernate

Prepared by: Database Architecture Team

Document Version: 1.0

Date: 29 July 2026

Classification: Confidential — Engineering Team Only

1. Introduction

This document is the Database Design Document (DDD) v1.0, Part-1: Database Foundation, for the School ERP System. It is the formal, standalone database design reference that Database Administrators and developers consult directly when implementing the physical PostgreSQL schema, consolidating the logical data model already established in Appendix-H, the database-specific engineering standards already fixed in Master Development Blueprint (MDB) Part-3, and the deployment-level database architecture already fixed in Appendix-L, Appendix-N, and MDB Part-4.

This document does not regenerate RGD v2.0, any prior Appendix, any UI/UX Screen Specification, or any MDB Part — every principle below is reused by reference, not re-derived. It contains no SQL, no CREATE TABLE statements, no entity class definitions, no ER diagram, and no sample data — it is a standards and conventions reference, not an executable or visual artifact.

Where MDB Part-3 already fixed a database standard (e.g., Entity Design Standards, Section 22), this document restates it at the level of detail a Database Administrator needs without contradicting it — Part-1 of this DDD is a formalization and consolidation of already-approved decisions, not a new design pass.

2. Database Objectives

- Provide the authoritative physical-design reference for all 39 entities already logically specified in Appendix-H, ensuring every table, column, key, and index a Database Administrator creates traces back to an approved logical element.
- Meet the Scalability and Performance NFR targets already fixed in Appendix-F (concurrent users, data volume, query response time), using PostgreSQL-specific mechanisms consistent with MDB Part-3 Section 21.
- Preserve full referential integrity, auditability, and soft-delete behavior uniformly across every table, exactly as already fixed in Appendix-H Sections 2.6-2.9 and MDB Part-3 Section 22 — no table is exempted from these baseline behaviors.
- Establish naming, typing, and indexing conventions precise enough that two different developers, or a developer and an AI coding assistant, produce structurally identical schema elements for the same logical entity without needing to consult each other.

3. Database Architecture Overview

Restates, without altering, the PostgreSQL architecture already fixed in MDB Part-3 Section 21 and the deployment topology already fixed in Appendix-L Sections 9-10 and Appendix-N Section 9.

Aspect	Standard (Reused)
Engine	PostgreSQL, single logical database per environment (MDB Part-1 Section 3)
Schema Organization	Single schema hosting all 39 entities, grouped by module for readability (MDB Part-3 Section 21) — not split into per-module schemas
Topology	Primary read/write instance plus at least one streaming-replication read replica (Appendix-L Section 9-10, MDB Part-3 Section 21)
Partitioning	Native declarative partitioning for AttendanceRecord, AuditLog, NotificationLog, Invoice/Payment (Appendix-H Section 17.8)
ORM Mapping	Spring Data JPA + Hibernate maps every entity per the Entity Design Standards already fixed in MDB Part-3 Section 22
Connection Management	HikariCP connection pooling per Application Layer instance (MDB Part-3 Section 21)

4. Database Design Principles

Consolidates the design principles already fixed across Appendix-H Section 2 and MDB Part-3 Sections 22 and 25 into a single reference list.

- Third Normal Form (3NF) is the default target for every table, with denormalization limited to documented JSON-typed columns holding variable-structure data (Appendix-H Section 2.1, Section 8 this document) — no other denormalization is permitted without a formal exception.
- Every table uses a single-column surrogate primary key; natural/business keys are retained only as unique-constrained candidate keys, never as the primary key (Appendix-H Section 2.3, Section 9 this document).
- No table supports a hard DELETE under normal application operation; every table implements the soft-delete pattern (Section 7 this document), consistent with Appendix-H Section 2.7.
- Every write-capable table carries a version column for optimistic-locking concurrency control (Appendix-H Section 2.8), with pessimistic locking reserved for the specific safety-critical cases already identified in MDB Part-3 Section 25 (e.g., seat-allocation capacity).
- Every session-scoped table carries an explicit academic_session_id foreign key rather than relying on date-range inference (Appendix-H Section 2.9) — this remains unchanged and is not reinterpreted at the physical level.
- Institute isolation (multi-tenancy) is not implemented at the physical schema level, consistent with the single-institute working assumption already fixed in Appendix-I Section 21.2 and Appendix-H Section 2.11 — the schema is not pre-emptively partitioned by institute.

5. Naming Standards

Consolidates the naming conventions already fixed in Appendix-H Section 2.2, Appendix-N Section 2.12, and MDB Part-1 Section 9 into the single definitive table for physical database objects.

Object	Convention	Example
Table	snake_case, plural, matches the entity name (Appendix-G/H)	students, invoices, marks_records
Column	snake_case, descriptive	admission_number, due_date
Primary Key Column	<table_singular>_id	student_id, invoice_id
Foreign Key Column	<referenced_table_singular>_id (or _ref_id for polymorphic, Section 9)	class_id, owner_ref_id
Primary Key Constraint	pk_<table>	pk_students
Foreign Key Constraint	fk_<table>_<referenced_table>	fk_invoices_students
Unique Constraint	uq_<table>_<column(s)>	uq_students_admission_number
Check Constraint	ck_<table>_<rule>	ck_marks_records_marks_range
Index	idx_<table>_<column(s)>	idx_invoices_student_id

6. Schema Standards

Fixes how the single PostgreSQL schema (Section 3) is internally organized, extending MDB Part-3 Section 21 with physical-design-level detail.

- Tables are logically grouped by the same 11-module structure already used throughout this documentation set (Admission, Student Information, Attendance, Examination, Fees, Library, Transport, HR & Payroll, Communication, Reports, Administration) — this grouping is a naming/documentation convention within the single schema, not a PostgreSQL schema-per-module split.
- Cross-cutting/system tables (User, Role, Guardian, AuditLog, Configuration, Document, ApprovalRequest — Appendix-H Section 4, Administration module) are treated as shared infrastructure tables, documented separately from the 10 business-domain module groupings.
- Junction tables resolving Many-to-Many relationships (RolePermission, ClassSubjectMap, StudentGuardianLink — Appendix-H Section 6) are documented alongside the two entities they connect, not as a standalone group.
- Lookup/reference tables already catalogued in Appendix-H Section 7 are grouped with their owning module (e.g., FeeHead with the Fee module) rather than in an undifferentiated "lookups" group, preserving traceability to their Appendix-G/H entity ID.

7. Common Columns Standards

Fixes the exact set of columns every table carries in addition to its entity-specific attributes, consolidating Appendix-H Sections 2.6-2.8 and MDB Part-3 Section 22 into one mandatory baseline.

Column	Purpose	Source Standard
<table>_id	Surrogate primary key (Section 9)	Appendix-H Section 2.3
version	Optimistic-locking concurrency control	Appendix-H Section 2.8, MDB Part-3 Section 25
created_by / created_at	Audit trail — who/when a record was created	Appendix-H Section 2.6, MDB Part-3 Section 22
updated_by / updated_at	Audit trail — who/when a record was last modified	Appendix-H Section 2.6, MDB Part-3 Section 22
is_deleted	Soft-delete flag; excluded from standard queries when true	Appendix-H Section 2.7
deleted_by / deleted_at	Audit trail for the soft-delete action itself	Appendix-H Section 2.7

These columns are populated automatically via Spring Data JPA Auditing (MDB Part-3 Section 22), never set manually in application code — this is a mandatory, non-optional baseline applied identically to all 39 entities, with no per-table exception.

8. Data Type Standards

Fixes the PostgreSQL data type used for each logical attribute type already catalogued in Appendix-G's Attribute Catalogue — this section maps logical types to physical types; it does not restate the attributes themselves.

Logical Type (Appendix-G)	PostgreSQL Data Type	Notes
Surrogate ID (BIGINT)	BIGINT (or BIGSERIAL/IDENTITY for generation)	Matches Appendix-H Section 2.3 exactly
VARCHAR(n)	VARCHAR(n)	Length n matches the Min/Max Length already fixed per attribute in Appendix-G
TEXT	TEXT	Used only for genuinely unbounded content (e.g., Circular.body), per Appendix-G
DATE	DATE	No time component, for date-only logical attributes (e.g., dob, attendance_date)
DATETIME	TIMESTAMP WITH TIME ZONE	Time-zone-aware, for all timestamp attributes (e.g., created_at, posted_at)
DECIMAL(p,s)	NUMERIC(p,s)	Exact precision required for all monetary and marks-related values — floating-point types are never used for money or marks
BOOLEAN	BOOLEAN	Direct mapping
JSON	JSONB	Binary JSON storage for the documented flexible-structure columns (salary_structure_json, deductions_json, grade_band_json, stops_json — Appendix-H Section 2.1)

9. Primary & Foreign Key Standards

Fixes physical key implementation, consolidating Appendix-H Sections 2.3-2.4 and 11 with MDB Part-3 Section 22.

- Every table's primary key is a single BIGINT column generated by a database identity/sequence mechanism — composite primary keys are used only for junction tables (Section 6), matching their composite-key definition already fixed in Appendix-H Section 6.
- Every foreign key is enforced as a database-level constraint referencing the parent table's primary key, with the default action RESTRICT on both update and delete — Cascade Delete is never used anywhere in the schema, consistent with Appendix-H Section 11 and MDB Part-3's soft-delete-only principle (Section 7 this document).
- Polymorphic associations (User.owner_ref_id, Document.owner_ref_id, ApprovalRequest, NotificationLog, BookIssue — Appendix-H Section 19) cannot be expressed as a single-target database foreign key; these columns carry no database-level FK constraint and rely on the application-layer integrity check already mandated in MDB Part-3 Section 22 — this is a documented, deliberate exception to the "every FK is a database constraint" rule above, not an oversight.
- Candidate/business keys (admission_number, invoice_no, etc.) are enforced via a unique constraint, never used as the primary key, consistent with Appendix-H Section 2.5.

10. Indexing Standards

Fixes physical index creation standards, consolidating Appendix-H Section 12 and MDB Part-3 Section 27.

- Every primary key receives its default unique index automatically; every foreign key column receives an explicit non-unique index, since PostgreSQL does not index foreign keys automatically.
- Every candidate/business key (Section 9) receives a unique index, matching the Unique Constraints already fixed per entity in Appendix-H Section 4.
- Composite indexes are created exactly where Appendix-H Section 12 already recommends them (e.g., AttendanceRecord's (student_id, timetable_entry_id, attendance_date)) — an index is not added speculatively for a column combination not already justified by a documented query pattern.
- Search-optimized indexes (Student.full_name, Book.title/author, Employee.full_name) are created to support the Search Performance target already fixed in Appendix-F NFR-SRCH-001, using PostgreSQL's text-search or trigram indexing capability rather than a plain B-tree index for free-text matching.
- Index creation is included in the same versioned migration standard already fixed in MDB Part-3 Section 28 — an index is never added ad hoc directly against a running database outside a reviewed migration.